



Company Data

Last Close	280.67
Market Cap	1.295T
Shares Outstanding	4,390,741,509
52-Wk High	301.50
52-Wk Low	172.00
EV/NTM EBITDA	10.4x
Fiscal Year End	31-Dec-2024

Valuation Results

DCF – Terminal Growth	335.34
DCF – Exit Multiple	346.23
Comps – EV/EBITDA	245.33
Comps – P/E	354.29
Target Price	367.26
Implied Return	34%

Financial Data

	21A	22A	23A	24A	25E
Revenue	52.7	58.7	79.4	90.5	94.1
Revenue Growth	4.6	11.3	35.2	13.9	4.0
Adj. EBITDA	26.3	49.9	76.2	88.8	97.8

Executive Summary

We are initiating coverage of **Contemporary Amperex Technology, (“CATL”)** with a BUY rating and a **\$351.10** 12M price target.

Our 12M price target at the date of coverage is USD 351.10. This was derived from a 50-50 blended of (1) relative valuation of trading comparable with the NTM EV/EBITDA and P/E multiples and (2) discounted cash flow.

CATL is a Chinese technology company best known for being the largest EV battery manufacturer. It supplies leading automakers and energy storage providers with premier lithium-ion and next-generation battery solutions. With strong foothold in the rapidly growing battery energy storage sector, *(thesis one should take like this amount)*, and *(thesis two should take like this amount)*, we view CATL as a business with immense growth potential and excellent competitive positioning.

Investment Thesis Highlights

Based on our DCF and comparable company analyses, we believe that CATL is an attractive investment opportunity offering a compelling risk-reward profile.

Thesis 1: CATL is in an advantageous position to capitalise on global battery energy storage system (BESS) growth

CATL is uniquely positioned to derive disproportionately outsized value from the rapidly expanding BESS space, which is primarily driven by the increasing take-up of renewable energies. Its unmatched resilient supply chain suggests that CATL can deliver lasting storage at scale, with others bogged down by bottlenecks caused by geopolitical headwinds. Additionally, alongside State Grid China’s large-scale infrastructure initiatives and aggressive investment in R&D, CATL’s product portfolio aligns well with global requirements of affordable, scalable and safe BESS solutions both now and in the future.

Thesis 2: CATL sustains EV battery leadership through scale, integration and global resilience

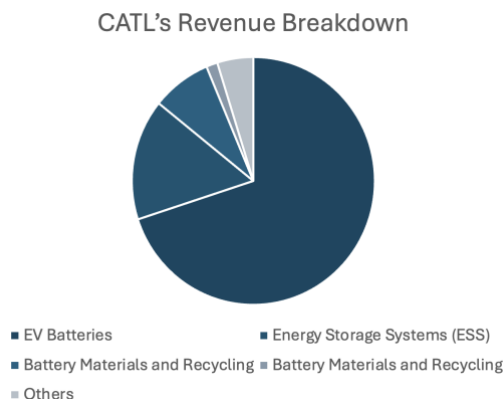
By controlling the full value chain—mining, recycling, cell-to-pack design, and localized gigafactories—CATL achieves 20–30% lower pack costs than peers, reinforced by platform innovations like CTP and the TENER Stack. Its ~38% global market share is locked in by multi-year OEM contracts and high switching costs, while new revenue streams such as BaaS and EVogo swapping expand recurring income opportunities. Meanwhile, advanced recycling and ESG-aligned operations insulate margins from raw material volatility and regulatory shocks, strengthening CATL’s resilience and reinforcing investor confidence in its long-term dominance.

CATL’s R&D Edge Drives Superior Innovation Returns and Margin Expansion

CATL’s unrivalled R&D productivity, exemplified by the industry’s first commercial sodium-ion battery, positions it years ahead of global competitors. Its ability to **rapidly scale lab breakthroughs into profitable mass production**, supported by strong state backing and cost competitiveness, has already secured a huge market in China and global customers such as Tesla. With **continuous R&D investment, robust cash reserves, and a proven commercialization track record**, CATL is

Adj. EBITDA Margin (%)	20.2	15.2	19.0	24.5	22.8
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Figure 1: CATL's Revenue Breakdown



Source: Company Filings

Figure 2: One of CATL's gigafactories, located in Germany



Source: Invest in Thüringia

well positioned to deliver superior returns on innovation, expand margins, and consolidate its dominance in the global battery sector.

Company Overview

Contemporary Amperex Technology Co., Limited (CATL) engages in the development, production and sale of energy storage batteries and systems worldwide. It offers EV batteries, such as battery cells, battery modules/racks and battery packs; ESS batteries comprise battery cells, battery racks, containers, and AC side systems; and battery material products, which include lithium compounds, precursors, and cathode materials, as well as battery recycling and after-sales services.

A leader in EV battery manufacturing, but also a pivotal player in powering the rapid shift to distributed energy worldwide

A global leader in EV battery manufacturing, CATL plays a pivotal role in powering the rapid shift to distributed energy worldwide. It is the largest producer of lithium-ion batteries for electric vehicles (EVs) and energy storage systems (ESS), commanding a market share of roughly 38%. CATL supplies leading international automakers, leveraging technologies that are often regarded as more advanced than those of its Western competitors. At the same time, CATL is expanding beyond mobility into grid and industrial applications through its fast-growing ESS business. In 2024, ESS accounted for 22.4% of CATL's total battery shipments, up from 19.4% in 2023, underscoring its importance as a growth driver. This segment positions CATL as not only a supplier of automotive batteries but also a comprehensive energy solutions provider.

CATL's ESS offerings span the entire electricity value chain:

- Power generation – stabilizing renewable output from solar and wind.
- Transmission and distribution – enabling grid frequency regulation, peak shaving, and intelligent load balancing.
- End-use consumption – supporting industrial and commercial facilities, residential energy storage, telecom base stations, uninterruptible power supply (UPS) systems, and off-grid applications.

By strengthening its foothold in both mobility and stationary energy, CATL is strategically positioned at the intersection of sustainable transport and the distributed energy revolution.

CATL is quickly developing global scale, anchored by innovation

CATL's strategic strength lies in the synergy between its global manufacturing scale and cutting-edge R&D capabilities. With more than a dozen gigafactories in China, and with its recent opening of advanced plants in Germany and Hungary, CATL is uniquely positioned to service both its domestic and key international markets. With its proximity to major automotive hubs, it can gain competitive advantages in cost efficiency, localised supply and stronger integration with global automakers.

On the innovation front, CATL invests heavily in chemistry and design breakthroughs: 1) Battery chemistries: leadership in Lithium Iron Phosphate (LFP), commercialising sodium-ion cells, and pursuing solid state, 2) Design architectures: pioneering Cell-to-Pack (CTP) and ultra-fast charging solutions like the Shenxing battery. 3) Energy Storage Systems (ESS): expanding from EVs into distributed energy, with products like EnerOne, EnerC, and TENER Stack, serving industrial, grid, and residential applications.

Figure 3: Increase in EV battery demand

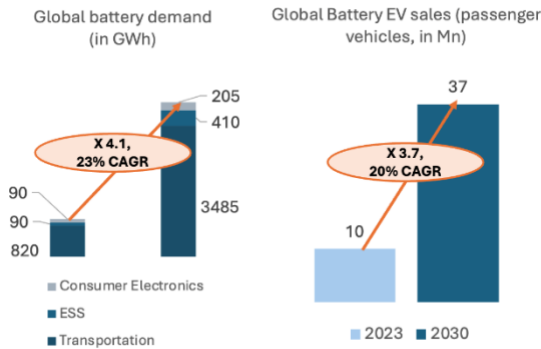
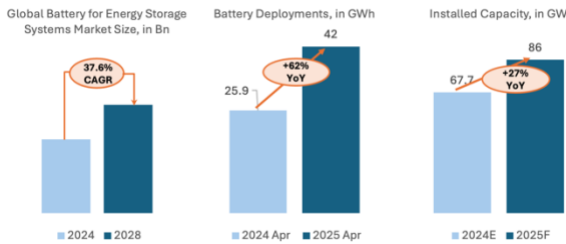
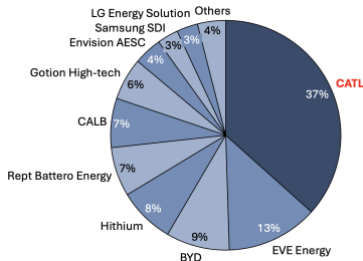


Figure 4: Global ESS Boom



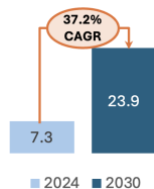
Source: ESS News, EnergyTrend, GlobalNewswire

Figure 5: ESS Market Share (2024)



Sources: Energy Storage News

Figure 6: Lithium-ion Battery Recycling Market Size in Bn



Source: McKinsey

Figure 7: Porter's Five Forces Analysis

Industry Summary

Untapped Growth Potential in Key Markets

Despite the sustained growth in the EV industry, there remains significant untapped potential in the global shift to electrification, with EV markets still expanding rapidly across major regions. Global EV sales surpassed 17million vehicles in 2024—20% of all new car sales—but the market is far from saturated, projected to reach 37 million units and 4100GWh of battery demand annually by 2030 (Figure 3). This sustained growth is driven not only in Europe, where EV penetration is expected to leap from 25% to 60% and import demand for batteries remains high, but also in fast-growing regions like Southeast Asia, where CATL is expanding its manufacturing footprint.

Explosive ESS Growth

The global Energy Storage System (ESS) market is experiencing exponential growth in 2025, propelled by surging demand for grid stability, renewable integration, and decarbonization initiatives. Total ESS market value is expected to reach \$29 billion this year, and this growth is set to continue: The global battery for ESS market size is estimated to grow by USD 47.19 billion from 2024-2028, with a CAGR of 37.62% during the forecast period. In addition, installed capacity saw a 27% year-on-year jump to 86GW and annual battery deployments up 62% to 42GWh (Figure 4). Major markets like China and the U.S. are hitting record installation milestones, and utility-scale as well as residential segments are both showing robust double-digit gains. CATL stands out as the global market leader, commanding a 36.5% share of ESS batteries and being recognized as the world's top producer in H1 2025 (Figure 5). This rapid expansion not only highlights ESS as a high-growth sector poised for further expansion but directly positions market leaders such as CATL to capitalize on powerful secular tailwinds, making them compelling investment candidates for long-term growth portfolios.

Stricter Regulations Accelerate Transition to Sustainable Battery Supply Chains

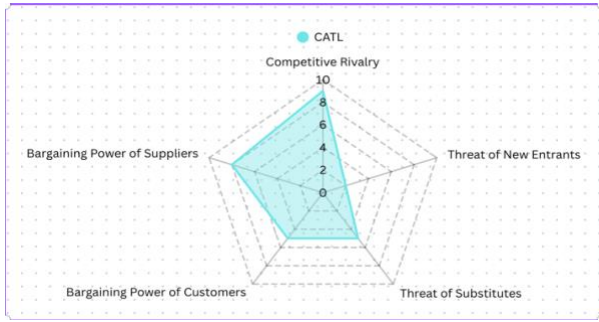
Globally, retired EV batteries are set to reach millions of tons within a decade, driving the recycling market to grow at 37.2% CAGR from 2025–2030 (Figure 6). In parallel, zero-carbon gigafactories powered by >80% renewables are becoming industry benchmarks, cutting millions of tons of CO₂ annually. Investment in recycling and sustainable manufacturing has surged into the tens of billions, as companies target carbon neutrality by 2030–2035. This shift is reinforced by regulation: the EU Batteries Regulation (effective 2025) mandates lithium battery recycling efficiency of 65% by 2025 and 70% by 2030, setting global standards that accelerate adoption of circular practices and reduced resource extraction.

Porter's Five Forces

CATL operates in a **highly competitive market**, facing intense rivalry from both global and domestic peers such as LG Energy Solution, BYD, and Panasonic. Overcapacity in China has further heightened price competition, pressuring margins, though CATL's scale, strong R&D capabilities, and established customer relationships provide it with a relative advantage.

The **threat of new entrants is low**, given the high capital intensity, technological complexity, and raw material requirements of the battery industry. However, government support in regions such as the U.S. and Europe for new capacity could gradually increase competition over time.

Supplier power is moderate, with CATL exposed to volatility in lithium, nickel, and cobalt markets. While these materials pose cost risks, CATL



Source: Team Estimates

has partially reduced its dependency through backward integration into mining projects and recycling operations.

Buyer power remains high, as global automakers are increasingly dual-sourcing batteries and investing in in-house capacity to reduce reliance on suppliers. This dynamic reinforces pricing pressure on CATL, though its technology leadership and innovation, including advanced chemistries like sodium-ion, help sustain demand.

The **threat of substitutes is moderate**, with alternatives such as solid-state batteries, hydrogen fuel cells, and other emerging chemistries under development but not yet commercially viable at scale. CATL’s investment in next-generation battery technologies acts as a hedge, though disruptive innovation remains a long-term risk.

Figure 8: Raw Material Security

CATL	High	Stake in Kisanfu copper-cobalt mine – direct control over cobalt supply
BYD	Medium	Heavier reliance on securing supplier agreements
LGES	Medium	Recent long-term offtake from Wescef’s Mt. Holland project with strong coverage in lithium but vulnerable to NA market compliance
Northvolt	Low	Vertical integration under development, no equity stakes in raw mineral projects

Sources: Straits Times, BYD, Reuters, Northvolt

Figure 9: Container Density (MWh/tonne)

Investment Summary

Investment Thesis 1: CATL is in an advantageous position to capitalise on global BESS growth

CATL’s valuation goes beyond its contribution of lithium ion-batteries to EVs and should also include its unrealised potential in its expansion into ESS facilities.

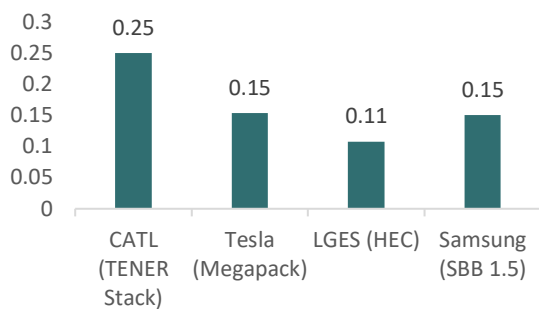
1. Raw material project involvement through equity stakes ensures supply chain robustness in volatile mineral markets

CATL’s current vertical integration involving heavy equity stakes in mineral acquisition, cathode and anode production, and pack assembly has contributed to a robust and resilient supply chain. Most notably, CATL had recently acquired a 30% stake in the Kisanfu high-grade copper-cobalt mine in the Democratic Republic of Congo, solidifying long-term contracts for access to one of the world’s largest cobalt deposits. As an extension to its upstream dominance, CATL’s position within CBC, a Chinese consortium, has also opened doors to up to 35,000 tons per annum of battery-grade lithium carbonate via the development of lithium extraction plants in Bolivian salt flats.

These upstream investments provide a leg-up over CATL’s peer battery players (Figure 8) who primarily rely on long-term supply contracts. This is especially critical when raw material costs comprise 70% of total BESS production costs, and where raw material price markets are highly volatile.

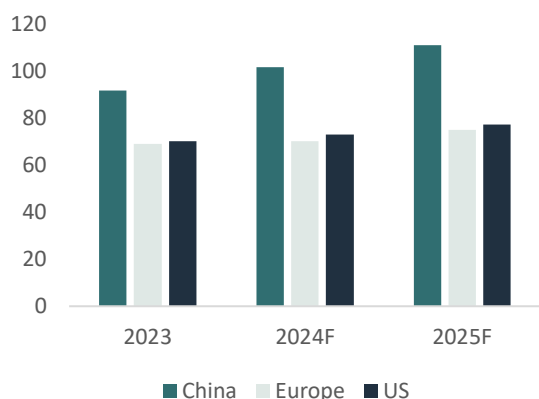
2. Long-term benefits for grid-scale ESS derived from the recent launch of the TENER Stack Energy Storage System

CATL’s recent technological innovation, the TENER Stack, is reported to gives a 45% improvement in volume utilisation and a 50% increase in energy density compared to traditional BESS systems (Figure 9). This space efficiency enables CATL to respond to more energy storage demands with a smaller footprint, making it extremely appealing to land-constrained developers. Furthermore, TENER Stack’s deployment



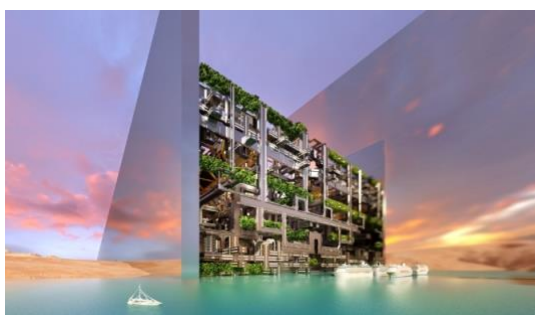
Sources: CATL, Tesla, LGES, Samsung SDI

Figure 10: Grid Capex, USD bn



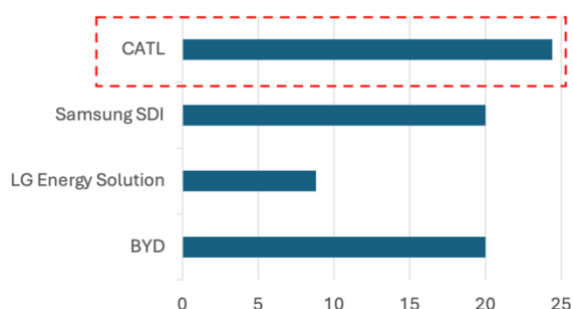
Source: Financial Times

Figure 11: Saudi Arabia's NEOM Project



Source: Tourist Saudi Arabia

Figure 12: Comparison of margins with industry competitors



flexibility – seen in its support of centralised and string Power Conversion System architectures – makes the system suitable for a wide variety of grid applications, signalling a competitive advantage.

TENER Stack's mobility also enhances CATL's technological leadership, as each unit weighs under 36 tonnes, thereby reducing shipping costs by up to 35%. LFP battery chemistry and cutting-edge safety technologies like advanced gas sensors also provide enhanced reliability and security.

This head start effectively creates more cost-efficient environments for its clients, making CATL the best positioned to serve a wider range of customers compared to its peers.

3. Support from Chinese State Grid Investments provides stronger, more sustainable demand

China's electrification expansion (Figure 10), driven by State Grid's investments in ultra-high-voltage infrastructure and energy storage creates a particularly favourable environment for CATL to capture outsized BESS demand. Through the state-owned electricity utility corporation, China plans to spend over USD100bn solely on upgrading grid infrastructure, with slightly over a tenth specifically set aside for ultra-high-voltage lines. These investments mark a shift towards renewable energy penetration, with increasing demand for low-carbon sources to generate as much as 50% of power within the next 5 years. Furthermore, in China, new-type energy storage capacities increased four-fold to 31.4GW, fuelled by a surge in lithium-ion deployments.

As the world's largest BESS battery supplier and China's industrial gem for energy storage solution, CATL is positioned to reap substantial benefits from its own state-led, grid-enabled BESS boom above its Western peers.

Investment Thesis 2: CATL sustains EV battery leadership through scale, integration and global resilience

1. Integrated Supply Chain Efficiency

CATL's controls across the entire value chain, from raw materials to pack design, enables it to secure the highest margins in the industry (Figure 12). CATL's mining stakes and closed loop recycling cut input costs by 10-20%, in-house cell/module manufacturing with >90% utilization delivers 5-10% cost savings via optimized production and economies of scale. Localized gigafactories reduce tariffs/logistics, bringing 5-10% savings. Furthermore, platform innovations such as cell-to-pack (CTP) and cell-to-rack (CRC) enhance structural efficiency and energy density. Notably, the TENER Stack energy storage platform improves volume utilization by 45%, increases energy density by 50% and reduces construction costs by 20%.

2. Market Share Lock-in & Revenue Visibility

CATL maintains a 38% global market share in H1 2025 underpinned by long-term contracts with large OEMs (Figure 13), unlocking durable and recurring revenue streams. High switching cost of EV batteries due to 2-3 years for requalification and >\$50 000 for design change create substantial lock-in and customer loyalty, reinforcing CATL's market position. CATL operates 13 global gigafactories fully compliant with local content rules in U.S./EU (Figure 14). CATL is also actively diversifying its revenue streams.

Battery-as-a-Service (BaaS) is an emerging revenue stream, with the global BaaS market projected to grow at a 32.4% CAGR, reaching approximately \$9.36 billion by 2033. CATL's EVogo battery swapping network further embeds customers into its ecosystem, fostering recurring service-based revenues and expanding customer touchpoints.

3. Resilient & ESG-Aligned Global Footprint

Source: Fitch Ratings, Tech Buzz China, Alinvest, Morningstar

Figure 14: CATL secures contracts with large OEMs



Figure 15: Map of CATL's R&D Centers and Production Bases

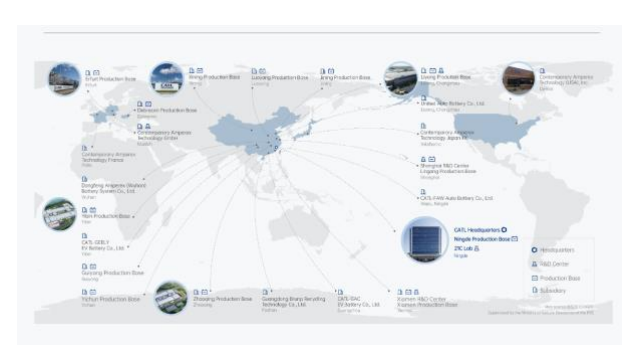


Figure 16: Comparison of Cell Performance

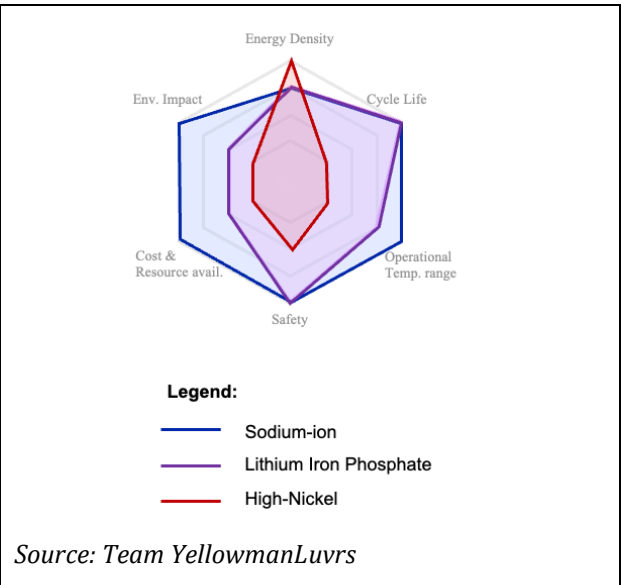


Figure 17: Chinese Government Subsidies to CATL (US\$, millions)

Closed-loop recycling system recovers up to 99.6% of nickel, cobalt, and magnesium, and 91% of lithium, insulating margins from raw material price fluctuations and driving sustainability leadership. Battery swapping and second-life battery programs extend asset utility and improve resource efficiency, and strong ESG performance translates to reduced financing costs and improved investor confidence, supporting CATL's innovation and growth. Gigafactories in Europe, U.S. and Indonesia also reduce exposure to policy shocks and tariffs, enhancing holistic supply chain resilience.

Investment Thesis 3: CATL's R&D Edge Drives Superior Innovation Returns and Margin Expansion

CATL's unrivalled R&D productivity, exemplified by the industry's first commercial sodium-ion battery, positions it years ahead of global competitors. Its ability to **rapidly scale lab breakthroughs into profitable mass production**, supported by strong state backing and cost competitiveness, has already secured a huge market in China and global customers such as Tesla. With **continuous R&D investment, robust cash reserves, and a proven commercialization track record**, CATL is well positioned to deliver superior returns on innovation, expand margins, and consolidate its dominance in the global battery sector.

1. Breakthrough in Sodium-Ion Batteries

Since 2023, sodium-ion battery technology has been **highly anticipated across the industry**, yet competitors have struggled to scale it beyond the lab. CATL broke through this bottleneck with the **April 2025 launch of its Naxtra sodium-ion battery**, becoming the first company worldwide to achieve commercial mass production. This positions CATL **one to two generations ahead of overseas peers**, many of whom have yet to achieve cost-effective LFP production (Annex 1). The Naxtra battery offers competitive **performance metrics**—175 Wh/kg energy density, 500 km driving range, over 10,000 charge cycles, and fast charging to 80% in 37 minutes—while also excelling in **safety** and leveraging the **resource abundance of sodium** to mitigate critical metal supply risks. (Figure 16) CATL has further strengthened its commercialization prospects by introducing a **hybrid battery pack** that integrates sodium-ion and lithium-ion cells, balancing sodium's low-temperature performance with lithium's higher energy density. With the sodium-ion market forecast to grow at a **27% CAGR from 10 GWh in 2025 to 70 GWh by 2033**, CATL is uniquely positioned to capture outsized share and extend its technology moat, making this a powerful catalyst for long-term value creation.

2. Rapid and Profitable Commercialization of R&D

CATL's vertically integrated supply chain and industry-leading cost base (as mentioned in Investment Thesis 2) enable it to **convert lab innovations into profitable mass-market products faster than peers**, driving superior R&D returns. (Figure 17) Strong state backing amplifies this advantage: In 2023, CATL received **US\$809 million in government subsidies — the highest of any company in China —** with 87.4% directed through sales tax exemptions, directly boosting adoption of new chemistries. With China accounting for ~70% of CATL's market, this policy environment provides both a **launchpad for domestic scaling** and a springboard for exports.

CATL's track record validates these advantages. In 2020, it was the **first to supply LFP at scale**, securing Tesla as a global customer. Demand exceeded supply, enabling CATL to negotiate **cost-plus contracts**, where lithium price fluctuations were passed to Tesla.

These advantages ensure CATL can **rapidly monetize new chemistries**—such as sodium-ion—turning high R&D spend into scalable revenue and durable margin expansion.

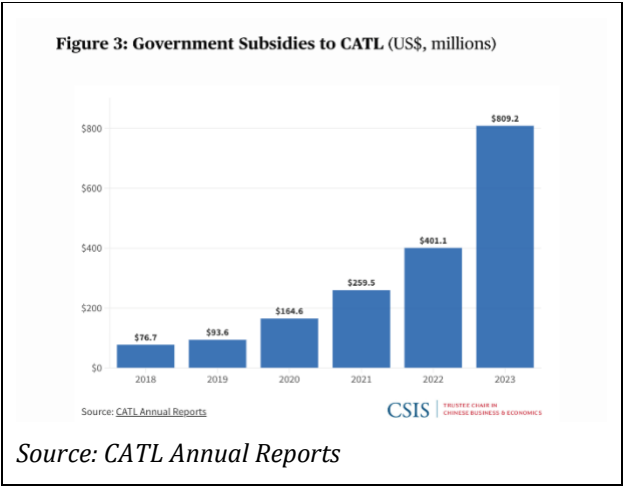


Table 1: ESG Scorecard

Provider	Approach / Focus	Overall ESG rating
LSEG	Scores based on 500+ metrics across E, S, and G pillars	78/100 (Excellent)
MSCI	Industry-relative risk and opportunity	AA (Leader)
Sustainalytics	Assesses ESG performance through Exposure (material ESG risk level) and Management (robustness of ESG programs)	Low Risk (19.7/100) Strong Management
WIND China	China-specific rating of A-share companies (AAA to C); focuses on local regulatory priorities like pollution control, energy efficiency, employee rights, and board governance.	8.15/10 (AA, 2023)

Figure 11: Comparison of GHG emission intensity with peers

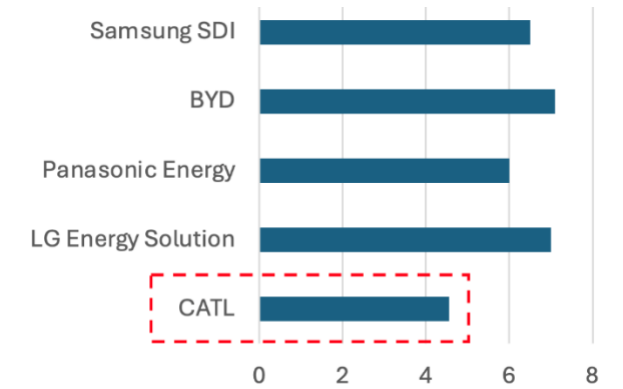


Figure 12: Comparison of renewable energy usage with peers

3. Healthy Operating Cashflow to Sustain Technological Leadership

Healthy operating cashflow is likely to reinforce this edge. CATL generated **¥97 billion in operating cash flow** in 2024 and held **over ¥300 billion in cash reserves** at year-end, providing flexibility to sustain high-intensity R&D and expand production capacity. This balance sheet strength also enables CATL to **bridge early cost gaps in scaling new products**, ensuring commercialization well before global competitors.

Combined with its proven ability to rapidly scale innovations—demonstrated by the mass adoption of LFP batteries and now the **world's first commercial sodium-ion launch**—CATL is positioned to **sustain long-term technological leadership, capture outsized global market share, and exercise potential price-setting power** in the battery sector.

ESG

With the increasing emphasis on sustainability, CATL's ESG profile significantly enhances stakeholder confidence in the business. CATL is consistently rated highly across major ESG agencies (Figure [xxx]) — **78/100 (Excellent) by LSEG, AA (Leader) by MSCI, Low Risk with Strong Management by Sustainalytics, and 8.15/10 (AA) by WIND China in 2023.** (Table 1) These scores place CATL well ahead of most industry peers, reflecting strong governance systems, robust risk management, and proactive sustainability integration.

Our internal analysis of the most material ESG factors for the battery industry indicates that CATL outperforms peers across all categories, with particular strength in governance and supply chain resilience. Unlike many competitors who face persistent risks around emissions disclosure and social practices, CATL's robust safeguards and disclosure quality mitigate these concerns. The company's consistent high ratings across global and domestic providers demonstrate both international credibility and strong alignment with China's regulatory ESG priorities.

Environmental

Greenhouse gas (GHG) emissions

In 2024, CATL's greenhouse gas emission intensity decreased by 21% year-on-year, reaching 4.56 million tons of CO2 equivalent per MWh produced, significantly outperforming industry peers (figure 11). This was driven by a 45% increase in distributed photovoltaic capacity to 350 GWh, with 74.51% of total power consumption sourced from zero-carbon electricity. Advanced AI-powered energy management systems and waste heat recovery further optimized production efficiency, resulting in a 15% reduction in energy consumption per GWh and minimized fossil fuel use.

Environmentally-friendly product design and innovations

CATL has extended the lifespan of its energy storage batteries to up to 18,000 charge cycles, helping to reduce material demand and lifecycle emissions. The company plans to deploy over 10,000 battery swap stations, improving battery utilization efficiency and enabling large-scale collection of retired batteries, supporting sustainability throughout the product lifecycle.

Recycling and Circular Economy

In 2024, CATL recycled 130,000 tons of end-of-life batteries, recovering about 17,000 tons of lithium salts. Operating the world's largest battery take-back and recycling network, CATL significantly reduces reliance on virgin raw materials and supports circular economy principles to minimize environmental impact.

Strategic Partnerships and Long-Term Circular Goals

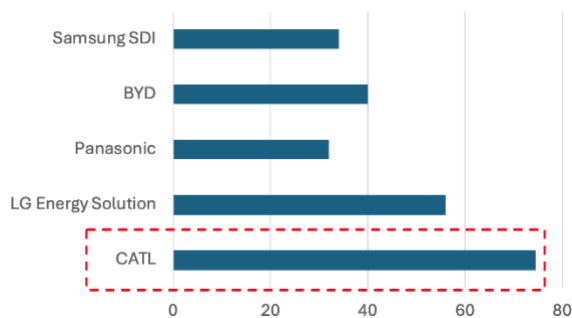
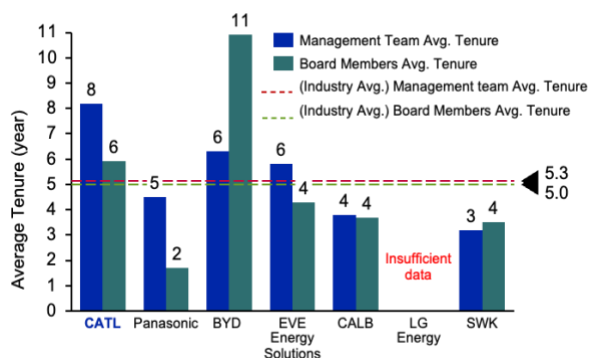


Figure 13: CATL's leadership tenure compared to peers



Source: Simply Wall Street

Figure 14: Ownership Structure

CATL collaborates with the Ellen MacArthur Foundation to integrate circular economy principles across the battery value chain. The partnership aims for 50% of new battery production to be decoupled from virgin raw materials within 20 years, marking a bold sustainability commitment for the industry.

Renewable Energy Leadership

CATL leads the battery industry in renewable energy usage, with the highest share of zero-carbon electricity powering its operations, reinforcing its position as an environmental leader among peers (Figure 12).

Social

CATL's ESG performance centers on strong commitments to employee welfare, community engagement, education and social responsibility.

Employee Welfare and Training: CATL has a workforce exceeding 130,000 employees globally as of 2025. The company invests significantly in employee training programs, with close to 60 hours of skill development and safety training annually for each employee.

Local Employment Impact: CATL's facilities in various regions aim to create thousands of jobs. For instance, the construction and operation of the Hungary plant alone are expected to generate around 9,000 direct and indirect jobs.

Community and Education Donations: CATL donated 110 million yuan to establish a primary school in Jiaocheng District in 2022. CATL's controlling shareholder's donation of shares worth 1.374 billion yuan to Shanghai Jiaotong University also supports long-term educational development and research capabilities.

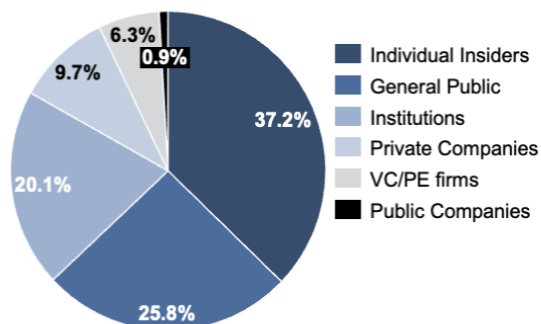
Governance

Leadership: CATL's governance is anchored by experienced leadership. The management team averages **8.2 years of tenure** and the board **5.9 years**, both above the industry average of **5.0 years** for management and **5.3 years** for boards (Figure 13). Founder, Chairman, and GM **Yuqin Zeng** has led the company since 2011 with **13.7 years** of continuous service, providing strategic continuity and stability that few peers can match.

Shareholder Ownership: Ownership structure (Fig. 14) further strengthens governance credibility. **Individual insiders hold 37.2% of shares**, the largest proportion, ensuring close alignment between management and shareholder interests. Within this, Founder Yuqun Zeng's **22.59% stake** reinforces accountability and underscores his long-term commitment to CATL's growth trajectory. The **general public (25.8%)** and **institutions (20.1%)** together provide both broad-based confidence and professional validation of the company's governance. Smaller holdings by **private companies (9.7%)**, **VC/PE firms (6.3%)**, and **public companies (0.9%)** add diversification while preserving insider-led stability.

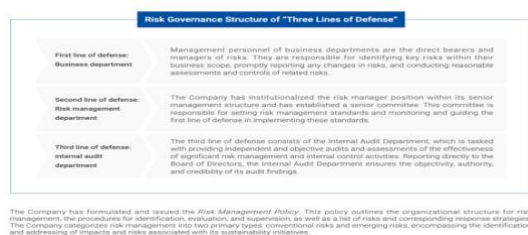
Governance safeguard: CATL embeds strong governance safeguards that mitigate operational and reputational risks. A **"Three lines of Defense" model**, a **strict anti-bribery framework**, and a **closed-loop crisis management system** covering prevention, response, and review ensure disciplined oversight and resilience. In 2024, CATL reported **zero major compliance breaches**, supported by **multi-layered data and supplier security protocols** that protect intellectual property and ensure continuity across the global supply chain.

Board Independence: While independent directors represent **less than half of the board (3 out of 8)**, this is offset by robust institutional safeguards, diverse director expertise, and significant insider



Source: Simply Wall Street

Figure 15: CATL's Three Lines of Defense



Source: CATL ESG Report 2024

Figure 16: WACC Computation

Weighted Average Cost of Capital	
Book Value of Borrowings	136,984,817,000
Market Value of Equity	1,201,306,876,862
Total	1,338,291,693,862
Weight of Debt	10.2%
Weight of Equity	89.8%
Cost of Debt	
Interest Expense	9611743000
Tax Rate	14.5%
Pre-Tax Cost of Debt	7.0%
After-Tax Cost of Debt	6.0%
Cost of Equity	
Risk-Free Rate (Rf)	4.23%
Equity Risk Premium	5.27%
Unlevered Beta	0.86
Cost of Equity (CAPM)	8.76%
Average WACC	8.5%

Source: Team YellowmanLuvrs

Figure 17: Comps Set

accountability through equity holdings. These mechanisms ensure that oversight remains effective despite slight imbalance.

Valuation

Valuation Price Target: **USD367.26**.

Our blended price target of USD 367.26 (34% upside) was calculated via a weighted average (50%/50%) of relative valuation (EV/EBITDA, P/E) and intrinsic valuation (DCF) methodologies. We assigned equal weightage to both methodologies to balance the benefits of market-based valuation with the fundamental approach of intrinsic valuation. The valuation ranges produced by different methods are visualized in the valuation football chart (figure 11).

Intrinsic Valuation

Our two-stage free cash flow to firm (FCFF) DCF computed an intrinsic valuation of USD 367.26 per share, representing a 34% upside. We chose a 5-year time horizon through FY30 to sufficiently encompass CATL's strategic initiatives, including technological advancements, service expansion, and potential acquisitions.

Revenue Build

For revenue, we forecasted revenue growth for each of CATL business segments, based on historical growth trends, management guidance as well as the industry demand drivers as listed in the investment theses, including CATL's potential to capture the growing BESS market.

Cost Build

Cost drivers were further broken down into cost of product sold, cost of services sold, SG&A and R&D expenses, which were projected as a percentage of revenue, we expect these costs to increase as CATL secures a greater share of the battery market, reflecting higher production volumes, expanded global operations and sustained investment in innovation required to maintain its competitive edge.

WACC

Cost of Equity was calculated using CAPM, beginning with the risk-free rate, assumed to be the US 10Y Treasury Yield of 4.23% (figure 12). We used Damodaran's US Market Risk Premium data to derive an Equity Risk Premium of 5.54%. We then multiply that by a 5-year regression unlevered beta of 0.86, obtained from CapIQ. Incorporating all these components, we arrive at 8.76% for our Cost of Equity.

The pre-tax cost of debt was adjusted for the tax shield, using CATL's effective tax rate over the past 5 years.

Terminal Growth Rate

Derived using the Gordon Growth Model, assuming a terminal growth rate of 2.5% - in line with long-term US GDP growth projections - and arrive at a terminal value of 432.55. This takes into consideration CATL's ability to maintain perpetual growth beyond the specified forecast period as well as discounting future cash flows conservatively to account for macroeconomic uncertainty. As such, our valuation framework reconciles near-term growth momentum with a reasonable long-term path.

Relative Valuation

Our relative valuation was based on NTM EV/EBITDA and P/E multiples, resulting in USD 662.02 implied share price.

Investment Recommendation

We recommend a BUY rating on CATL with a target price of USD 367.26 and an entry price of USD 273.60, reflecting an 34% upside over a 12-month investment horizon. Our bullish outlook is driven by strong revenue visibility from the CATL's strong position in both the R&D segment and the support provided by its extensive vertical integration, which provides higher margins and the maintenance of longer-term

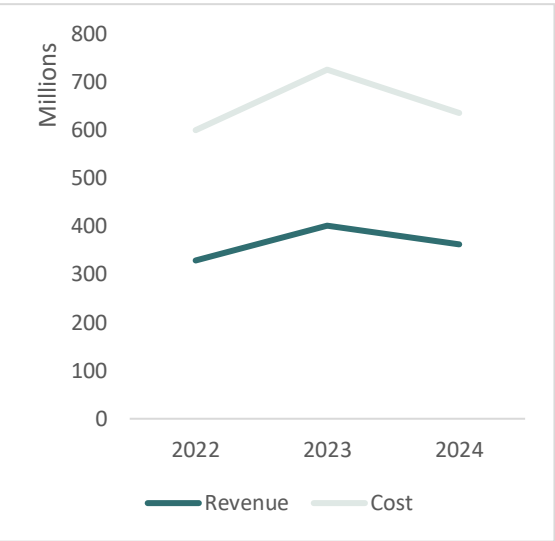
			
			
			

Source: Team Estimates

competitive edge, offsetting potential cyclical weaknesses in the macroeconomic sphere. Additionally, artificial intelligence tailwinds and eco transition trends support long-term demand for CATL's battery solutions.

Key catalysts include market share gains in high-growth markets supported by green transition policies, increased adoption of electric vehicles, and operational efficiencies enhancing profitability.

Figure 14: Profitability ('000 RMB)



Source: Fintech Ratings

Figure 15: Liquidity

Financial Analysis

Profitability

CATL's high ROE reflects effective capital use, likely driven by global scale, cost control, and operational efficiency. The widening gap between revenue and costs signals improving profitability and the very strong Q2 2025 gross margin—particularly in recycling and materials—shows strategic diversification enhancing earnings resilience amid competitive pressure.

CATL's profitability is underpinned by economies of scale, vertical integration, and technology leadership.

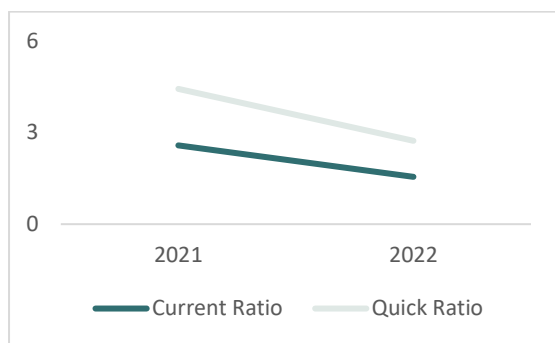
Liquidity

Liquidity remains healthy, supported by robust operating cash flows and a strong cash position that leaves the company in a net cash state, with reserves exceeding total debt by over RMB 160 billion. This liquidity buffer ensures CATL can meet short-term obligations while continuing to fund expansion. Although its current and quick ratios have moderated from their historical peaks—standing at roughly 1.5 and 1.2 in 2022—these levels still indicate comfortable short-term solvency (FIGURE 15).

Solvency

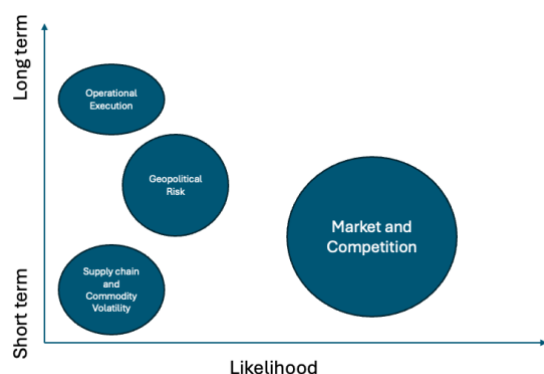
From a solvency perspective, CATL maintains a conservative capital structure despite rapid growth. Its debt-to-equity ratio of roughly 0.65 and historical asset-liability ratio in the 45–58% range suggest a moderate reliance on leverage, well within safe bounds. Crucially, CATL's net debt remains negative, indicating that it holds more cash than total borrowings, whichh provides significant balance sheet flexibility.

External confidence is reflected in its reaffirmed A- rating, with Fitch citing robust profitability and cash flow generation as key supports for long-term credit quality. In addition, CATL's diversified customer base across China, Europe, and North America reduces concentration risks,



Source: Templates Hub

Figure 21: Risk Analysis



Note: size of circle indicates severity of risk.

Sour

ensuring more stable cash inflows to service obligations. Strategic financing choices—balancing retained earnings, equity issuance, and measured debt—have also reinforced solvency. Together, these factors underpin a solid long-term financial foundation, allowing CATL to sustain its global expansion without compromising its credit profile.

Efficiency

The increase in receivables turnover and fall in inventory turnover suggests supply chain and working capital efficiency is shifting positively. This shows that there is strong demand for CATL's batteries, and inventory management as well as working capital management is in good order, amplifying our confidence in its financial health.

Investment Risks and Mitigation

Geopolitical risk

Amid escalating US-China tensions, CATL faces heightened scrutiny and regulatory barriers such as tariffs, quotas, and local content requirements, disrupting operations and limiting expansion.

Mitigant

Localization of Production: CATL has established gigafactories in Europe, Indonesia, and the U.S. to align with local content rules, avoid tariffs, and sidestep regional trade barriers. This supports compliance and reduces exposure to geopolitical shocks and helps meet local demand.

Revenue Diversification: CATL lowers its reliance on the U.S. market, focusing on Europe, Southeast Asia, and new markets. This helps cushion against trade restrictions and geopolitical escalations. Localization helps secure long-term contracts under local rules and insulate the company from sudden trade restrictions.

Market and Competition

CATL's dominance is threatened by fierce competition from rivals like BYD and LG Energy Solution. Margin compression from ongoing EV price wars and rapid innovation among competitors could erode market share or profitability.

Mitigant

Research & Development and Breakthrough Technologies: CATL invests heavily in R&D—over US\$2.5 billion in 2023—with a focus on breakthrough technologies like sodium-ion and ultra-fast charging batteries. This innovation, combined with AI-driven research and a vast patent portfolio, helps CATL maintain a competitive edge, protect margins, and fend off rivals despite intense market competition and margin pressure.

Vertical Integration: CATL controls multiple stages of the supply chain—from raw material sourcing to production and recycling—helping reduce costs, stabilize supply, and enhance bargaining power.

Supply Chain and Commodity Volatility

Fluctuations in critical mineral prices (lithium, nickel, cobalt) and potential disruptions in global supply chains may substantially impact input costs and margins.

Mitigant

Closed-Loop Recycling: Provides a stable, sustainable source of critical minerals by recovering and reusing materials from end-of-life batteries, significantly reducing exposure to price swings and external supply shocks.

Long-Term Supply Agreements: Locks in prices and ensures reliable access to raw materials, smoothing out market volatility.

Advanced Materials R&D: Developing alternative battery chemistries (like sodium-ion) decreases reliance on scarce minerals and adds flexibility to the sourcing strategy.

Appendix:

Appendix 1: Comparison of Cell Characteristics

Type of Cell	Energy Density	Cycle Life	Operational temp. range	Safety	Cost & Resource Availability	Environmental Impact	Overall Performance
Sodium-ion	Good 185Wh/kg (expected to exceed 200Wh/kg in the future)	Excellent 6000-20000	Excellent -40°C to +70°C	Excellent Non-flammable, non-susceptible to explosions /short circuits	Excellent \$40-\$70/kWh Na is the 5 th most abundant element in Earth's crust ; ~60,000× more conc. in seawater than Li	Low 20% less CO2 emission than Li-based cells; uses abundant materials; minimal water consumption Easiest to recycle due to non-hazardous nature. Can be fully discharged.	Legend: - Sodium-ion - Lithium Iron Phosphate - High-Nickel
Lithium Iron Phosphate (LFP)	Good 195Wh/kg	Excellent 2000-5000	Good -20°C to +50°C	Excellent Excellent thermal stability	Moderate \$80-\$100/kWh Iron and phosphate are non-toxic, abundant, though Lithium may become scarce in the future.	Moderate Iron and phosphate are non-toxic, abundant. Lithium, however, requires mining, Low intrinsic toxicity, but less incentive for recycling.	
High-nickel (NMC)	Excellent 220Wh/kg	Poor 500-1500	Poor 0°C to +45°C	Moderate 80% higher fire risk than LFP	Poor \$120-\$150/kWh Relies on nickel and cobalt, which are resource-limited and expensive	High Intensive mining of nickel/cobalt needed. Highest CO2 and toxic waste However, valuable metals incentivize recycling	